



**Department of Electrical and Computer
Engineering**
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Lab 9 : “PWM”

Introduction and objectives

Pulse Width Modulation (PWM) is a widely used technique in embedded systems for controlling power delivered to electrical devices such as motors, LEDs, buzzers,...

By rapidly switching an output pin between high and low states with varying duty cycles, PWM allows efficient and precise control of these devices using digital outputs.

MC9S12DT256 includes 6 PWM channels that make generating and managing PWM signals straightforward.

In this lab, you will learn how to initialize PWM channels, set appropriate clock sources and prescalers, control duty cycles and configure polarity to achieve desired output waveforms on the buzzer.

Pre Lab9 :

In this preliminary task, you are required to answer the following questions:

1. Locate the buzzer on the appropriate schematic diagram and find its pin connection with the MCU.
2. By referring to the 'PWM module' datasheet (located in 'Datasheets/PWM' folder), explain the role of the following registers:
 - PWME

- PWMCLK
 - PWMPRCLK
 - PWMPOL
 - PWMCTL (*bit4 CON01*)
 - PWMPER
 - PWMDTY
 - PWMCNT
3. Explain in details the process of generating a PWM signal by referring to the datasheet 'PWM module' (*last page*).
discuss the ideas related to : clock selection , PWMPER and PWMDTY registers, comparison with PWMCNT , Toggling or restarting the signal.

Lab 9 : “Buzzer”

Write fully commented programs for the PBMCUSLK board, including appropriate labels, memory operands and constants, to solve the 2 tasks below:

Task 1:

The first program should do the following:

Create a 1Khz signal on channel PWM0 with a duty cycle of 50% and send it to the Buzzer.

Task 2:

The second program should do the following:

- a. Create a 587hz signal with a duty cycle of 70% on both channels :
 - PWM0 , must be connected to the buzzer
 - PWM1, must be connected to an LED.
- b. Modify the previous program and let the duty cycle of PWM1 signal be 30%.
- c. Deduce the effect of the duty cycle on the brightness of the LED.
